Diaphragm Pump Design & Simulation Workflow

Engineering Design Steps, Inputs & Simulation Pipeline

1. REQUIREMENTS & PERFORMANCE (Initial Sizing)

Key Inputs:

- Required Flow Rate Q
- Maximum Operating Pressure P_{max}
- Vacuum / Suction Conditions $P_{vac/suc}$

Analytical Calculations:

- **Displacement Volume:** $V_d = \frac{Q}{N \cdot \eta_v}$
- **Stroke & Eccentric Radius:** $S = 2r \quad , \quad V_d = A_{eff} \cdot S$
- **Motor Power:** $P_{motor} = \frac{Q \cdot \Delta P}{\eta_{total}}$
- **Shaft Diameter:** $d \geq \left(\frac{16T}{\pi \tau_{allow}}\right)^{1/3}$
- **Valve Port Area:** $A_v = \frac{Q}{v_{max}}$

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Table 1.1 Gas Medium - Pump Function Focus

Function	Key Design Parameter	Key Design Equation	Main Focus
Compression	Pressure discharge maximization	$P_{motor} = \frac{\gamma}{\gamma - 1} \frac{P_{in} \cdot Q}{\eta} \left[\left(\frac{P_{out}}{P_{in}} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right]$	Diaphragm stress under high pressure & thermodynamic gas compression
Vacuum Generation	Minimum Dead Volume (V_{dead})	$P_{vac} \approx \frac{V_{dead}}{V_d + V_{dead}} \cdot P_{atm}$	Check valve opening & sealing under extremely low delta-P
Transfer	Maximum Flow Rate (Q) with minimum pressure drop	$Q = V_d \cdot N \cdot \eta_v$	Internal flow optimization & port pressure drop reduction
Aspiration	Suction Pressure & Impact Resistance	$A_v = \frac{Q}{v_{max}} \quad ,$ $\Delta P = \xi \cdot \frac{\rho v^2}{2}$	Diaphragm structural integrity against pulse loading & vacuum deflection

Table 1.2 Liquid Medium - Pump Function Focus

Function	Key Design Parameter	Key Design Equation	Main Focus
Transfer	Continuous Flow Rate (Q) & Incompressibility	$P_{hydraulic} = Q \cdot \Delta P \quad , \quad P_{motor} = \frac{Q \cdot \Delta P}{\eta_{mech} \cdot \eta_{vol}}$	Laminar/turbulent flow in valve passages & continuous hydraulic load
Aspiration	NPSHR & Cavitation Prevention	$NPSHA = \frac{P_{atm} - P_{vapor}}{\rho g} - h_s - h_f > NPSHR$	Cavitation detection during suction stroke & dynamic response of inlet valve
Dosing	Stroke Volume Repeatability & Response Time	$V_{dose} = V_d = A_{eff} \cdot S \quad , \quad \text{Error} = \frac{\lvert V_{real} - V_{target} \rvert}{V_{target}} \leq 1\%$	Transient stroke displacement accuracy & valve sealing delay during dosing pulse
Metering	Flow Linearity & High Precision Control	$Q(N) = A_{eff} \cdot S \cdot N \cdot \eta_v(P_{back})$	Volumetric efficiency vs. backpressure & high-precision velocity profiles in CFD

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2. MATERIAL SELECTION & TRANSIENT STRUCTURAL

Inputs:

- Material Properties (EPDM, NBR, PTFE - Hyperelastic models)
- Boundary Conditions (Fixed outer rim, center deflection/force)

- **Hyperelastic Strain Energy Potential:** $W = W(I_1, I_2, I_3)$
- **Stress-Strain Relations:** $S_{ij} = \frac{\partial W}{\partial E_{ij}}$
- **Fatigue Life Prediction:** $N_f = C \cdot (\Delta\sigma)^{-\beta}$

Outputs / Evaluation:

- **Diaphragm Deflection Profile**
- Stress & Strain Distribution & Fatigue Life Analysis

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3. FLUID DYNAMICS (CFD)

Inputs:

- Fluid Properties (Density ρ, Viscosity μ)
- Diaphragm Deflection Profile from FSI: Dynamic / Moving Mesh – UDF in ANSYS FLUENT

Formulas & Governing Equations:

- **Navier-Stokes Equations:** $\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}\right) = -\nabla P + \mu \nabla^2 \vec{v} + \vec{f}$
- **Continuity Equation:** $\nabla \cdot \vec{v} = 0$
- **Valve Pressure Drop:** $\Delta P_{valve} = \xi \cdot \frac{\rho v^2}{2}$

Outputs / Evaluation:

- Valve Pressure Drop & Flow Velocity Profiles
- Volumetric Efficiency Verification

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4. SYSTEM-LEVEL SIMULATION

Input:

- Calculated motion profile for Signal Generator in Simulink

Block Components:

- **Electric Motor Equations:** $T_m = K_t \cdot I \quad , \quad V = IR + L\frac{dI}{dt} + K_e \omega$
- **Kinematic Linkage / Crank-Slider:** $x(t) = r\cos(\omega t) + \sqrt{l^2 - r^2 \sin^2(\omega t)}$
- **Check Valve Dynamics:** $m\ddot{x}_v + c\dot{x}_v + kx_v = A_v \Delta P(t)$
- **Instantaneous Mass Flow Rate:** $\dot{m}(t) = \rho \cdot A(t) \cdot \sqrt{\frac{2\Delta P(t)}{\rho}}$

Final Outputs:

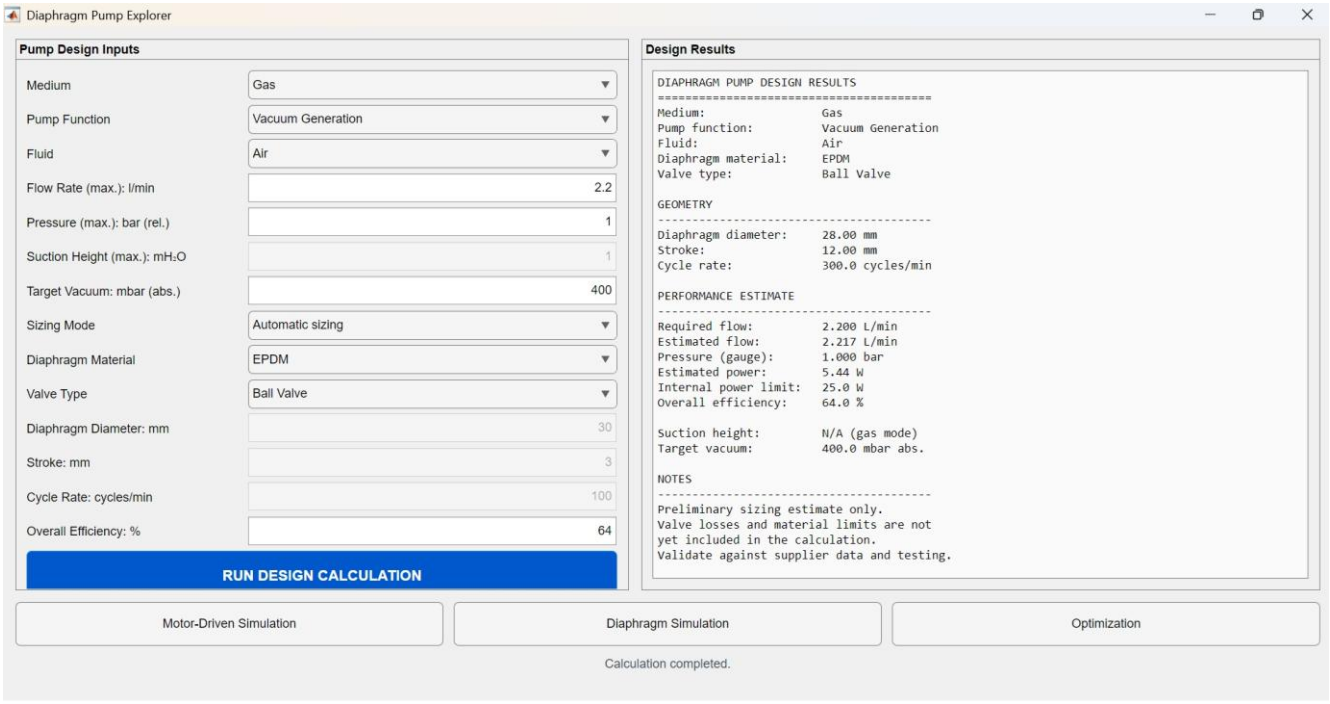
- Characterization Curves
- Overall System Performance Optimization

Diaphragm Pump Design & Simulation Platform

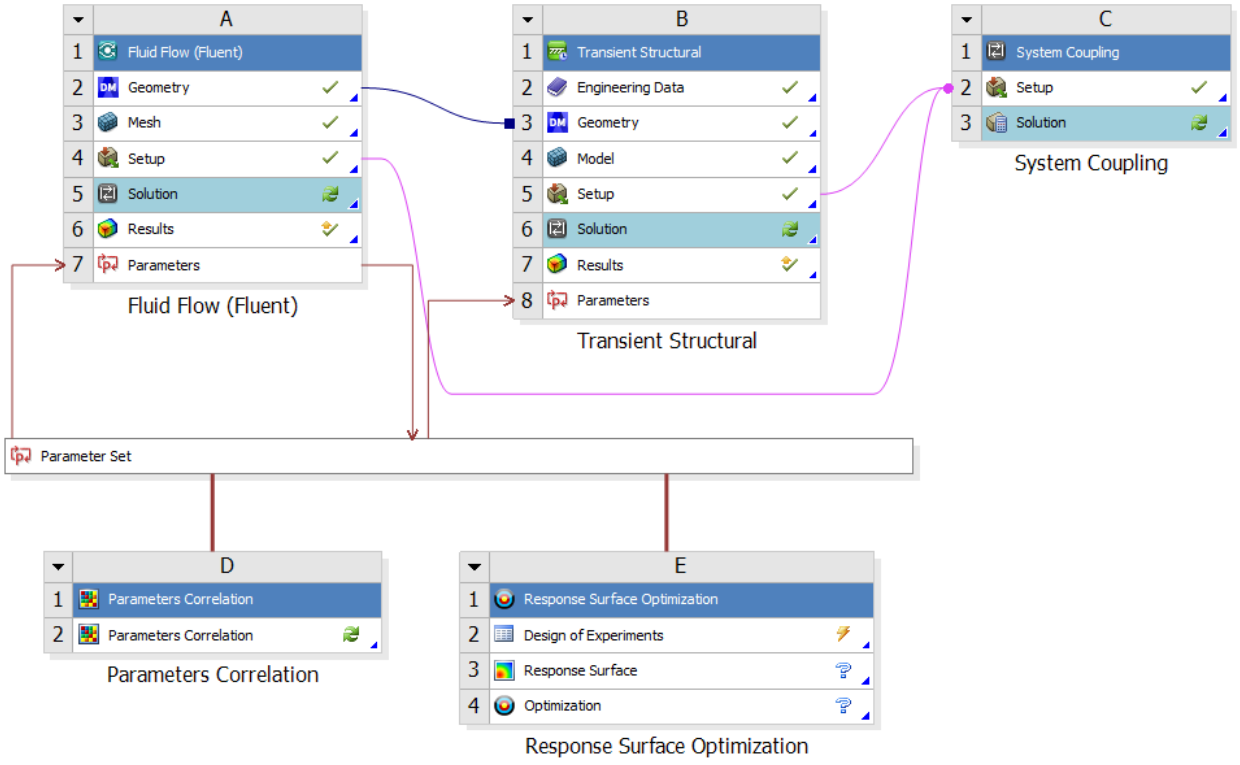
An interactive engineering platform for the conceptual design, preliminary sizing, simulation, and optimization of diaphragm pumps for gas and liquid handling across diverse laboratory applications.

Tools: MATLAB (GUI Development) | Simulink / Simscape (System-Level Simulation) | ANSYS (Component-Level Simulation)

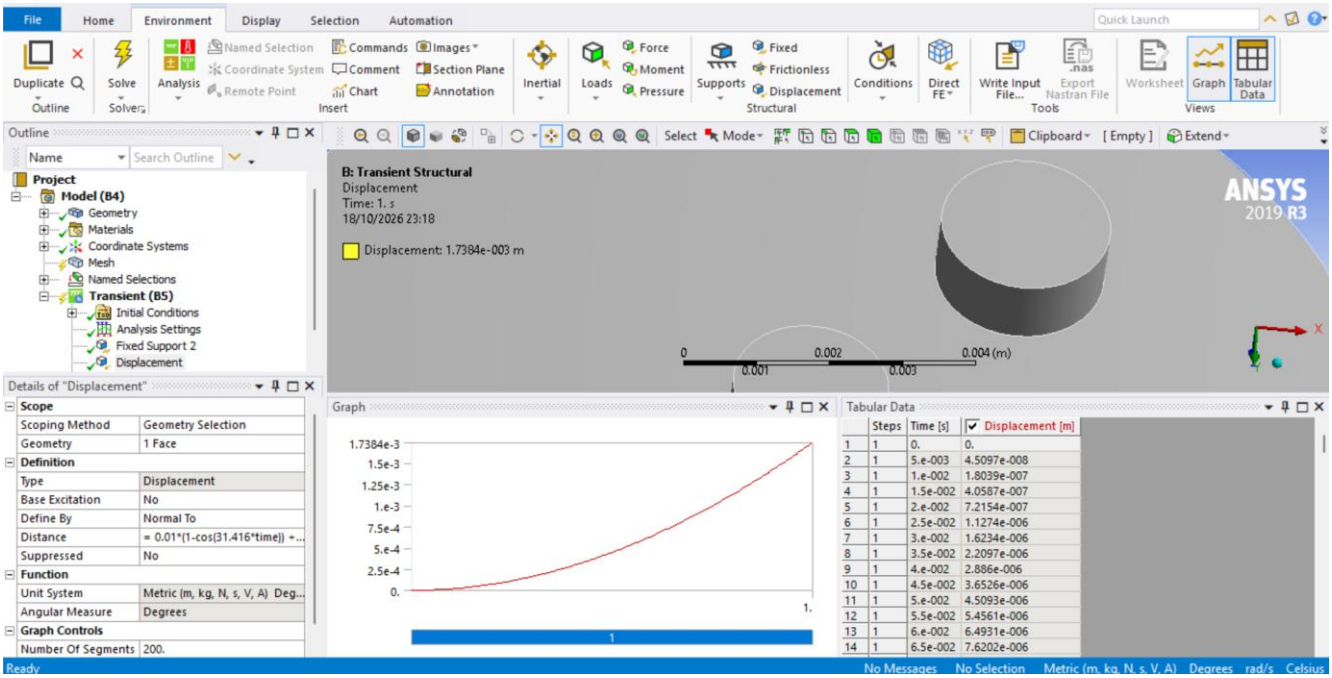
Feature	Gas Pump Platform	Liquid Pump Platform
Medium	Gas	Liquid
Functions	Aspiration, Compression, Transfer, Vacuum Generation	Aspiration, Dosing, Metering, Transfer
Design Inputs	Flow Rate, Outlet Pressure, Vacuum Level	Flow Rate, Pressure, Suction Height
Diaphragm	Diameter, Stroke, Material	Diameter, Stroke, Material
Valve	Valve Type, Valve Material	Valve Type, Valve Material
Sizing	Automatic / Manual	Automatic / Manual
Performance Analysis	Flow Rate, Pressure, Vacuum	Flow Rate, Pressure, Dosing Performance
Simulation	Motor-Driven & Diaphragm Motion	Motor-Driven & Diaphragm Motion
Optimization	Geometry & Operating Parameters	Geometry & Operating Parameters
Output	Preliminary Design & Performance Results	Preliminary Design & Performance Results



ANSYS Simulation – CFD Coupled with Structural



Displacement Profile – ANSYS Transient Structural

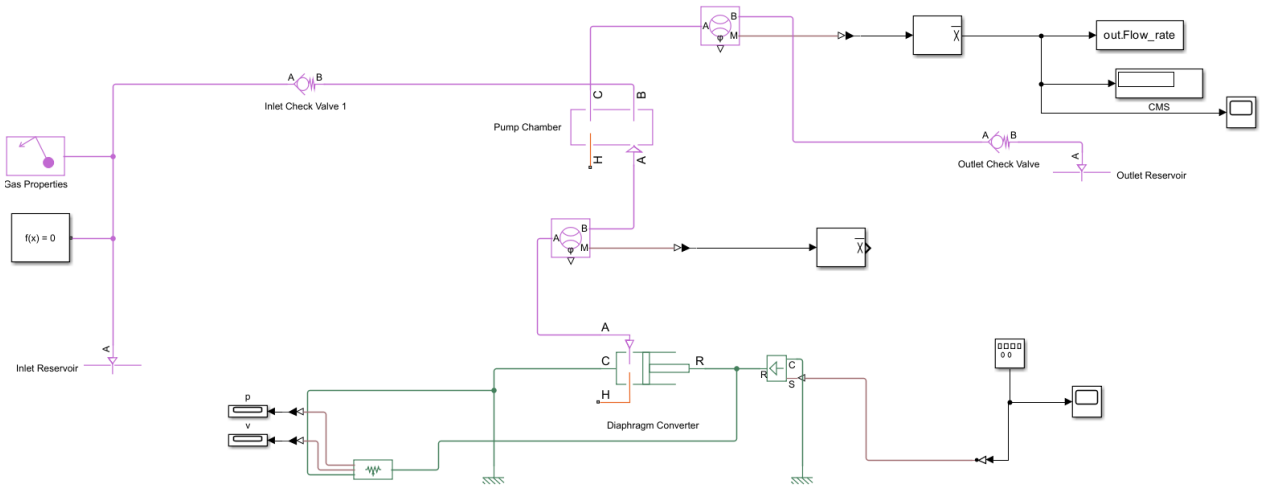


Parameters Definition – ANSYS Workbench

Outline of All Parameters

	A	B	C	D
1	ID	Parameter Name	Value	Unit
2	Input Parameters			
3	Fluid Flow (Fluent) (A1)			
4	P1	Chamber_R	0.008	m
5	P2	Chamber_Depth	0.004	m
6	P7	Inlet_Port_R	0.0010693	m
7	P8	Outlet_Port_R	0.0011618	m
8	P14	Diaphragm_Thickness	0.0008	m
9	Transient Structural (B1)			
10	P9	Analysis Settings Stiffness Coefficient	0.0005	
11	P10	Analysis Settings Mass Coefficient	0.32	
12	P11	Crank_R	0.0081	
13	P12	Rod_L	0.0104	
14	P13	RPM	900	
*	New input parameter	New name	New expression	
16	Output Parameters			
17	Fluid Flow (Fluent) (A1)			
18	P15	Out_MassFlowrate	7.1125E-05	m^3 s^-1
*	New output parameter		New expression	
20	Charts			

System Simulation without Motor – MATLAB Simulink



System Simulation with Motor RPM function – MATLAB Simulink

